

PROJECT REPORT

COMICCRAFT:

AI-POWERED COMIC STORY GENERATOR

A Mini Project Report

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ABSTRACT

ComicCraft is a web-based application developed to generate personalized comic stories and illustrations using Artificial Intelligence. The system allows a user to provide a story prompt together with details such as character name, setting, tone, and preferred art style.

The application combines Google's Gemini models for structured story and dialogue generation with Stable Diffusion for comic-style image generation. FastAPI provides the backend, while HTML, CSS, and Jinja2 templates support the user interface. The generated content is arranged into a five-panel comic and can be previewed before being exported as a PDF.

The project demonstrates how Generative AI can combine text generation, image synthesis, layout management, and document generation within a single workflow. ComicCraft is intended as a simple platform for students, storytellers, and creative users who want to turn ideas into visual stories.

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1. INTRODUCTION

1.1 Introduction

Artificial Intelligence has transformed digital content creation by enabling computers to generate text, images, and other forms of media. Comic creation traditionally requires a combination of writing, illustration, character design, panel arrangement, and final formatting.

ComicCraft is designed to simplify this process through a single web application. It combines Google's Gemini models for natural-language story generation with Stable Diffusion for image synthesis. The application accepts user-defined story details and converts them into a structured comic.

FastAPI is used for the backend, while HTML, CSS, and Jinja2 templates provide the web interface. The result is a workflow in which a user's idea is converted into story content, panel illustrations, a comic preview, and a downloadable PDF.

1.2 Scope of the Project

The scope of ComicCraft is to provide a web-based platform for generating personalized comics using Artificial Intelligence. The project focuses on integrating text generation, AI image creation, comic layout management, preview, and PDF generation into one workflow.

The current version supports single comic generation and does not include user authentication or cloud storage. Its modular structure allows these capabilities to be considered for future enhancement.

1.3 Motivation

Traditional comic production can require artistic skills, creative writing ability, and significant time. Many users may have story ideas but lack the skills or resources needed to turn those ideas into illustrated comics. ComicCraft is motivated by the goal of making this creative process simpler and more accessible through Generative AI.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

Creating a comic normally involves writing a script, designing characters, illustrating scenes, arranging panels, and formatting the final output. AI tools for text and images are often available separately. ComicCraft addresses the need for a unified workflow that combines these stages into one application.

2.2 Objectives

- Design a simple interface for collecting user inputs.
- Generate a structured comic outline using Gemini Flash.
- Generate detailed narration and dialogues using Gemini Pro.
- Generate comic-style illustrations using Stable Diffusion.
- Organize generated content into a structured comic layout.
- Provide a preview before downloading.
- Export the completed comic as a professionally formatted PDF.

2.3 PROJECT FEATURES

- AI-based comic story generation from user prompts.
- Customization of character name, setting, tone, and art style.
- Automatic generation of five-panel comic stories.
- AI-generated illustrations for each panel.
- Dynamic comic preview using Jinja2 templates.
- PDF export for offline viewing and sharing.
- FastAPI-based backend with a responsive web interface.
- Modular architecture for maintenance and future enhancements.

3. TECHNOLOGY STACK

Technology	Role	Purpose in ComicCraft
Python	Programming language	Core application logic and AI workflow
FastAPI	Backend framework	Handles requests, routing, and generation workflow
Google Gemini Flash	AI text model	Creates the structured comic outline
Google Gemini Pro	AI text model	Generates detailed narration and dialogue
Stable Diffusion	Image generation model	Creates comic-style panel illustrations
HTML / CSS	Frontend	Builds the user interface and presentation
Jinja2	Template engine	Renders dynamic comic preview pages
PDF generation	Output	Creates downloadable comic documents

3.1 Software Requirements

- Python environment for application development.
- FastAPI and related Python packages.
- Google Gemini API access for text generation.
- Stable Diffusion or configured image-generation service.
- HTML/CSS/Jinja2 templates.
- PDF generation library/tool used by the project.
- Modern web browser for accessing the application.

3.2 Hardware Requirements

- Standard personal computer or laptop.
- Keyboard and mouse.
- Stable internet connection for AI service requests.
- Hardware capable of running the selected development environment.

3.3 Why These Technologies Were Chosen

The technologies were selected for their suitability for AI-assisted web application development. Gemini models support structured narrative generation, Stable Diffusion supports image synthesis, FastAPI provides a lightweight backend, and Jinja2 supports dynamic presentation of generated content. Together they form a modular workflow from user input to final comic output.

4. SYSTEM ARCHITECTURE AND PROJECT WORKFLOW

4.1 System Architecture

ComicCraft follows a modular web-application architecture. The user interacts with the frontend through a browser. The FastAPI backend receives the input and coordinates the AI generation process. Text generation and image generation are handled by their respective AI services, after which the generated content is assembled into a comic layout and prepared for PDF export.

- User Interface Layer – collects story prompt, character, setting, tone, and art-style information.
- FastAPI Backend Layer – receives requests and controls the application workflow.
- Story Generation Layer – uses Gemini models to create the comic structure, narration, and dialogue.
- Image Generation Layer – uses Stable Diffusion to create panel illustrations.
- Layout Layer – combines text and images into the comic presentation.
- PDF Export Layer – prepares the final comic as a downloadable document.

4.2 Project Workflow

1. User enters the story prompt and customization details.
2. FastAPI receives and validates the request.
3. Gemini Flash generates the structured comic outline.
4. Gemini Pro generates detailed narration and dialogue.
5. Stable Diffusion generates an illustration for each panel.
6. The application combines text and images into a five-panel layout.
7. The comic is displayed for preview.
8. The completed comic is exported as a PDF.

4.3 Data Flow

The main data flow begins with user input and ends with the generated comic. Input data is passed from the browser to the FastAPI backend. The backend sends appropriate prompts to the text-generation models and receives structured story information. Panel descriptions are then used for image generation. The resulting text and images are passed to the layout stage, where they are rendered for preview and PDF export.

The modular data flow makes it possible to update individual components without redesigning the entire application.

5. MODULE DESCRIPTION

5.1 Overview

ComicCraft is divided into modules so that each major task can be developed and maintained independently. The modules work together to convert a user idea into a completed comic.

5.2 User Interface Module

The user interface collects the story prompt and customization details such as character name, setting, story tone, and preferred art style. It provides a simple form-based interaction and displays the generated comic preview.

5.3 Backend Module – FastAPI

The FastAPI backend manages requests, validates inputs, coordinates calls to the AI services, handles the generation workflow, and connects the frontend with the application logic.

5.4 Story Generation Module

The story-generation module uses Gemini Flash to create a structured comic outline and Gemini Pro to produce detailed narration and dialogues. The generated content is organized so that each panel has the information required for the next stage.

5.5 Image Generation Module

The image-generation module converts panel descriptions into comic-style illustrations using Stable Diffusion. A separate illustration is generated for each panel according to the story content and selected art style.

5.6 Layout Builder Module

The layout builder combines the generated narration, dialogue, and panel illustrations into a structured five-panel comic. It manages the visual arrangement so that the generated material can be previewed as a complete comic.

5.7 PDF Export Module

The PDF export module converts the completed comic layout into a downloadable document. This allows the generated comic to be saved for offline viewing and sharing.

5.8 Module Interaction

The modules operate sequentially: the interface collects input, FastAPI coordinates processing, the story module generates content, the image module creates illustrations, the layout module assembles the comic, and the PDF module produces the final output.

6. IMPLEMENTATION AND RESULTS

6.1 Implementation Overview

The ComicCraft implementation connects the frontend, FastAPI backend, AI generation services, layout process, and PDF export into one workflow. The system accepts user-defined story information and processes it through the configured generation modules.

6.2 Home Page

The home page acts as the entry point to the application. It provides the fields required to describe the comic idea and select customization options.

6.3 Comic Generation Process

After the user submits the form, the backend begins the generation process. The story is structured, detailed text is produced, and panel illustrations are generated. The outputs are then passed to the layout stage.

6.4 Comic Preview

The preview stage displays the generated panels together with their corresponding story content. This gives the user an opportunity to view the completed comic before exporting it.

6.5 PDF Export

After preview, the comic can be exported as a PDF. The export stage combines the generated content into a final document that can be viewed or shared outside the application.

6.6 Results

The project demonstrates an end-to-end Generative AI workflow in which a textual idea is transformed into a multi-panel visual comic. The described implementation successfully combines story generation, image synthesis, layout management, preview, and PDF generation within one web application.

7. ADVANTAGES

- Reduces the effort required for manual comic creation.
- Combines text and image generation in one workflow.
- Allows basic customization of generated stories.
- Provides a visual preview before export.
- Produces a portable PDF output.
- Uses a modular design that supports future development.

7.1 LIMITATIONS

- The current version supports single comic generation.
- User authentication is not included.
- Cloud storage is not included.
- Generation quality depends on the configured AI models and prompts.
- AI service availability and response time can affect generation.

7.2 FUTURE SCOPE

- User login and authentication.
- Cloud storage for saving generated comics.
- Multilingual comic generation.
- Additional themes and visual styles.
- Voice input for story creation.
- Real-time editing of generated comics.
- Mobile application versions.
- Improved AI models and generation quality.

7.3 CONCLUSION

ComicCraft demonstrates the practical use of Generative Artificial Intelligence in creative storytelling. By integrating Gemini-based text generation, Stable Diffusion image generation, FastAPI backend processing, dynamic web templates, comic layout management, and PDF export, the project provides a complete workflow from a user's idea to a finished comic.

The project also demonstrates how different AI and web technologies can be combined into a modular application. Although the current implementation has limitations such as the absence of authentication and cloud storage, the architecture provides a foundation for future enhancements.

8. REFERENCES

- Project source documentation and implementation materials.
- Google Gemini model documentation referenced by the project.
- FastAPI framework documentation.
- Stable Diffusion documentation and model resources.
- Python documentation.
- Jinja2 template documentation.